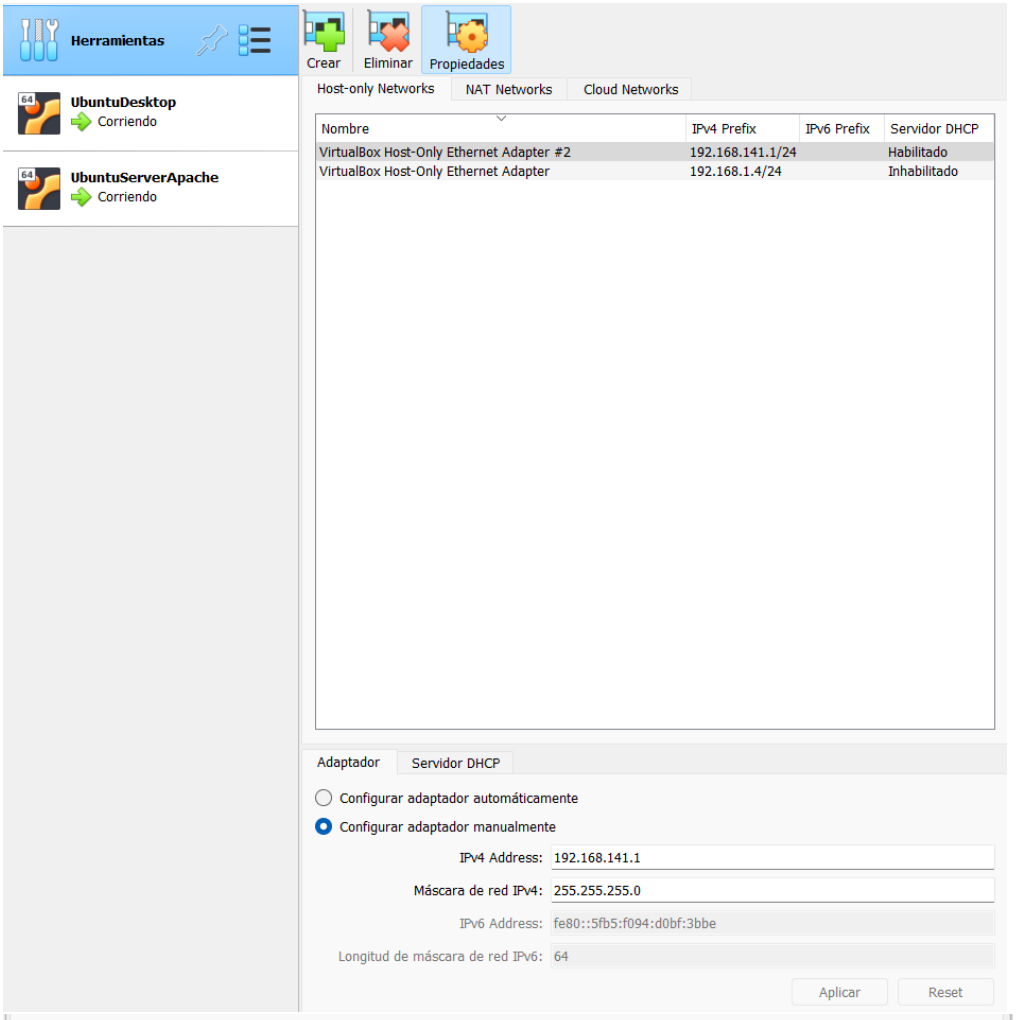
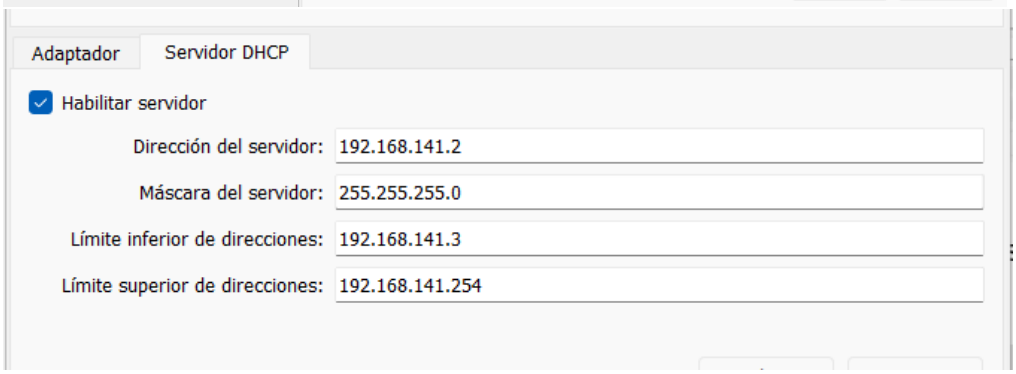
UD2: EJERCICIO DE VIRTUAL HOST

1. Vamos a configurar el Ubuntu Server de nuestro Virtual Box para que tenga dos interfaces de red: Uno en modo Host-only Networking (adaptador solo anfitrión) y otro como Red NAT (configurada en las preferencias, red 10.0.2.0/24 por ejemplo).





Nombre	IPv4 Prefix	IPv6 Prefix	Servidor DHCP
NatNetwork	10.0.2.0/24	fd17:625c:f037:2::/64	Habilitado

General Options | Reenvío de puertos

Nombre: NatNetwork

IPv4 Prefix: 10.0.2.0/24

Enable DHCP

Enable IPv6

IPv6 Prefix: fd17:625c:f037:2::/64

UbuntuServerApache
Corriendo



Sistema

Memoria base: 2048 MB
Orden de arranque: Disquete, Óptica, Disco duro
Aceleración: Paginación anidada, Paravirtualización KVM

Pantalla

Memoria de vídeo: 16 MB
Controlador gráfico: VMSVGA
Servidor de escritorio remoto: Inhabilitado
Grabación: Inhabilitado

Almacenamiento

Controlador: IDE
IDE secundario maestro: [Unidad óptica] Vacío
Controlador: SATA
Puerto SATA 0: UbuntuServerApache.vdi (Normal, 8,00 GB)

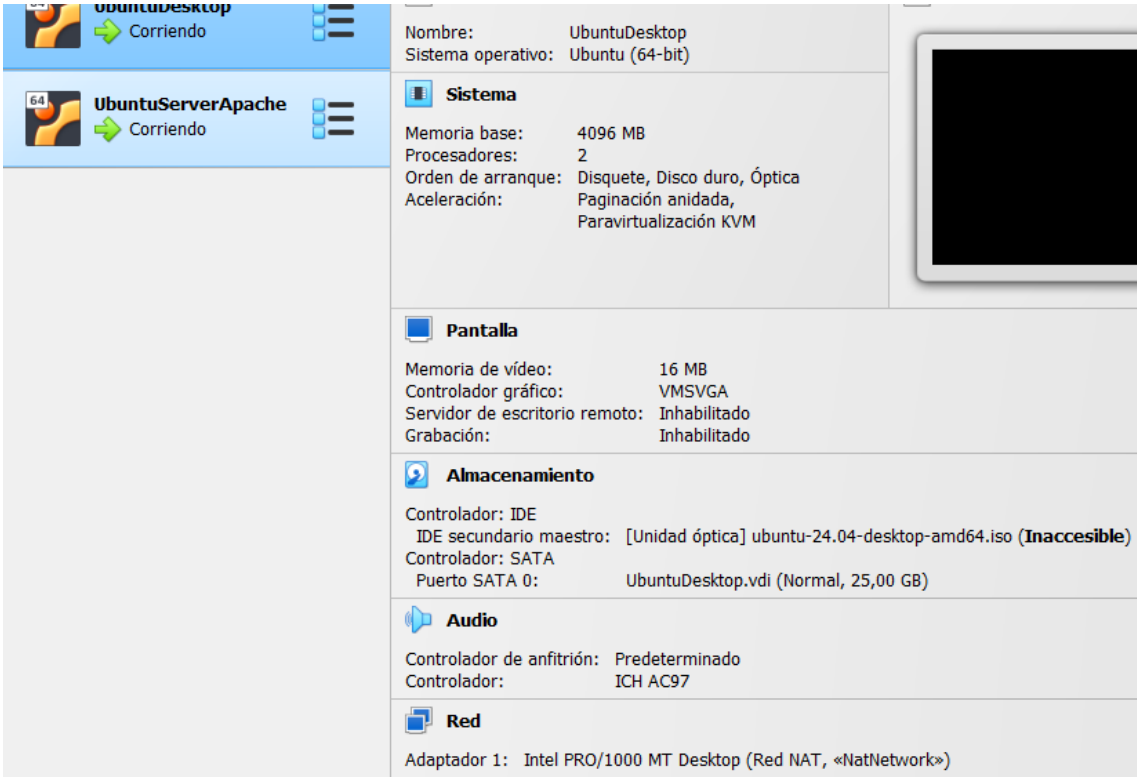
Audio

Controlador de anfitrión: Predeterminado
Controlador: ICH AC97

Red

Adaptador 1: Intel PRO/1000 MT Desktop (Adaptador solo anfitrión, «VirtualBox Host-Only Ethernet Adapter #2»)
Adaptador 2: Intel PRO/1000 MT Desktop (Red NAT, «NatNetwork»)

2. Instalamos un Ubuntu Desktop en otra máquina virtual que tenga solamente Red NAT.



3. Cuando esté bien configurado, vuestro Windows se podrá comunicar con el Ubuntu Server, y éste con vuestro Windows y con el Ubuntu Desktop (comprobadlo con pings)

Server

```
root@serverapache:/Lorea/comun# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.141.3  netmask 255.255.255.0  broadcast 192.168.141.255
        inet6 fe80::a00:27ff:fe48:afdd  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:48:af:dd  txqueuelen 1000  (Ethernet)
        RX packets 449  bytes 53184 (53.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 303  bytes 23136 (23.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.9  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe92:53e8  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:92:53:e8  txqueuelen 1000  (Ethernet)
        RX packets 47455  bytes 71561476 (71.5 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 12023  bytes 740537 (740.5 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 344  bytes 29966 (29.9 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 344  bytes 29966 (29.9 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Desktop

```
root@ubuntudesktop:/home/lorea# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.6  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe75:1f0  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:75:01:f0  txqueuelen 1000  (Ethernet)
        RX packets 240  bytes 305058 (305.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 164  bytes 17113 (17.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Bucle local)
        RX packets 188  bytes 18431 (18.4 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 188  bytes 18431 (18.4 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

Ping Desktop-Server

```
root@ubuntudesktop:/home/lorea# ping -c4 10.0.2.9
PING 10.0.2.9 (10.0.2.9) 56(84) bytes of data.
64 bytes from 10.0.2.9: icmp_seq=1 ttl=64 time=0.870 ms
64 bytes from 10.0.2.9: icmp_seq=2 ttl=64 time=0.599 ms
64 bytes from 10.0.2.9: icmp_seq=3 ttl=64 time=0.785 ms
64 bytes from 10.0.2.9: icmp_seq=4 ttl=64 time=0.690 ms

--- 10.0.2.9 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.599/0.736/0.870/0.101 ms
root@ubuntudesktop:/home/lorea# nano /etc/hosts
```

Ping Server-Desktop

```
root@serverapache:/Lorea/comun# ping -c4 10.0.2.6
PING 10.0.2.6 (10.0.2.6) 56(84) bytes of data.
64 bytes from 10.0.2.6: icmp_seq=1 ttl=64 time=0.606 ms
64 bytes from 10.0.2.6: icmp_seq=2 ttl=64 time=0.491 ms
64 bytes from 10.0.2.6: icmp_seq=3 ttl=64 time=0.968 ms
64 bytes from 10.0.2.6: icmp_seq=4 ttl=64 time=1.21 ms

--- 10.0.2.6 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3076ms
rtt min/avg/max/mdev = 0.491/0.819/1.213/0.287 ms
```

Ping Windows-Server

```
C:\Users\lorea>ping 192.168.141.3

Haciendo ping a 192.168.141.3 con 32 bytes de datos:
Respuesta desde 192.168.141.3: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.141.3: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.141.3: bytes=32 tiempo<1m TTL=64
Respuesta desde 192.168.141.3: bytes=32 tiempo<1m TTL=64

Estadísticas de ping para 192.168.141.3:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 0ms, Máximo = 0ms, Media = 0ms
```

Ping Server-Windows

```
rtt min/avg/max/mdev = 0.491/0.819/1.213/0.287 ms
root@serverapache:/Lorea/comun# ping -c4 192.168.14.33
PING 192.168.14.33 (192.168.14.33) 56(84) bytes of data.
64 bytes from 192.168.14.33: icmp_seq=1 ttl=127 time=1.74 ms
64 bytes from 192.168.14.33: icmp_seq=2 ttl=127 time=1.66 ms
64 bytes from 192.168.14.33: icmp_seq=3 ttl=127 time=1.58 ms
64 bytes from 192.168.14.33: icmp_seq=4 ttl=127 time=1.43 ms

--- 192.168.14.33 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.432/1.602/1.738/0.113 ms
```

4. En el servidor web el contenido de cada virtual host irá en una carpeta dentro del directorio **/"vuestroNombre"**. Mirad un ejemplo en apache2.conf de cómo hacerlo.

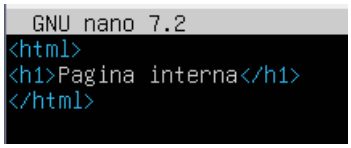
   mkdir /Lorea
   mkdir /Lorea/comun
   mkdir /Lorea/interna
   mkdir /Lorea/externa

5. Tendremos 3 virtual host: interna, externa, común. El contenido de las mismas es irrelevante, pero tienen que diferenciarse para saber a cuál estamos accediendo.

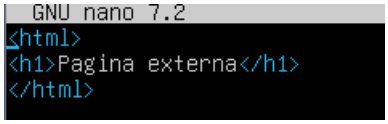
   nano /Lorea/comun/index.html

```
  GNU nano 7.2
<html>
<h1>Pagina comun</h1>
</html>
```

nano /Lorea/interna/index.html

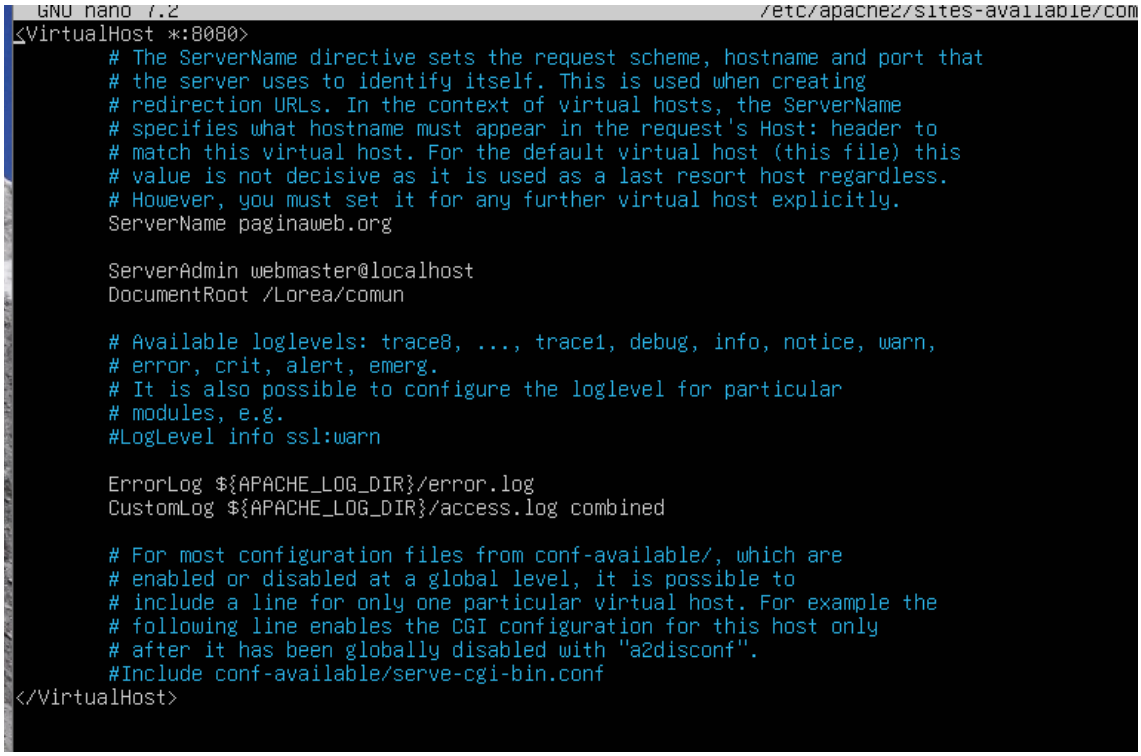


nano /Lorea/externa/index.html



6. **interna** solamente será accesible desde la interfaz de la red interna en el puerto 80, es decir, desde el Ubuntu Desktop. **externa** solamente será accesible desde vuestro Windows y el puerto 80. **común** será accesible desde ambas máquinas, pero desde el puerto 8080 solamente

nano /etc/apache2/sites-available/comun.conf



nano /etc/apache2/sites-available/interna.conf

```
 GNU nano 7.2                                          /etc/apache2/sites-available/in
<VirtualHost 10.0.2.9:5500>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        ServerName paginaweb.org

        ServerAdmin webmaster@localhost
        DocumentRoot /Lorea/interna

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

nano /etc/apache2/sites-available/externa.conf

```
 GNU nano 7.2                                          /etc/apache2/sites-availabl
<VirtualHost 192.168.141.3:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        ServerName paginaweb.org

        ServerAdmin webmaster@localhost
        DocumentRoot /Lorea/externa

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>
```

Habilitar los hosts virtuales
cd /etc/apache2/sites-available
a2ensite comun
a2ensite interna
a2ensite externa

Deshabilitar el default
a2dissite 000-default.conf

Modificar el archivo de configuración de apache2

nano /etc/apache2/apache2.conf

```
<Directory />
        Options FollowSymLinks
        AllowOverride None
        Require all denied
</Directory>

<Directory /usr/share>
        AllowOverride None
        Require all granted
</Directory>

<Directory /var/www/>
        Options Indexes FollowSymLinks
        AllowOverride None
        Require all granted
</Directory>

<Directory /Lorea/>
        Options Indexes FollowSymLinks
        AllowOverride None
        Require all granted
</Directory>

#<Directory /Lorea/comun>
#       Requier ip 192.168.141
#</Directory>
```

Modificar el archivo de configuración de ports

nano /etc/apache2/ports.conf

```
#Listen 80
Listen 192.168.141.3:80
Listen 10.0.2.9:5500
Listen 8080

<IfModule ssl_module>
        Listen 443
</IfModule>

<IfModule mod_gnutls.c>
        Listen 443
</IfModule>
```

Reiniciar Apache2

systemctl restart apache2

7. A las páginas (a todas) se accederá escribiendo en el navegador **paginaweb.org** (y el puerto si fuera necesario).

   Desktop-etik frogak:

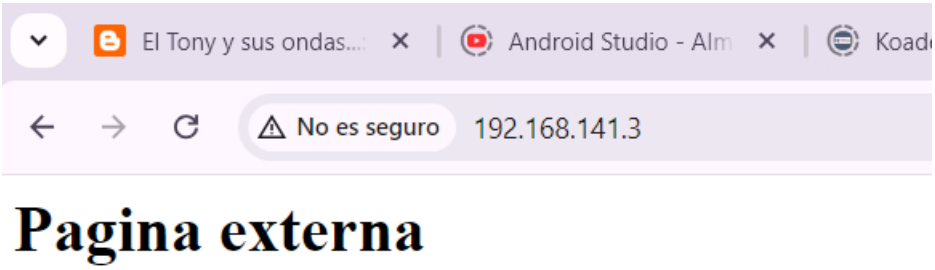
   Pagina Comun



   Pagina Interna



   Desde Windows



www.paginaweb.org???

8. Prueba la funcionalidad de la opción **Indexes** para el virtual host **común**.

¿¿¿Si en el servidor desaparece el archivo de index.html q pasaría???Para realizar esta prueba:

cd /Lorea/comun

Crear dos fitxeros cualquiera

nano Fitxategi.txxt
nano Fitxategi2.txt

ls

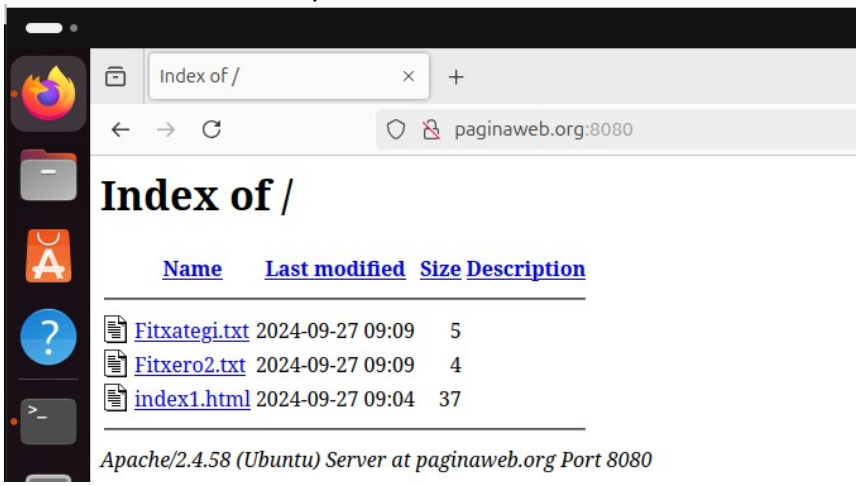
```
root@serverapache:/etc/apache2/sites-avail
root@serverapache:/Lorea/comun# ls
Fitxategi.txt  Fitxero2.txt  index.html
root@serverapache:/Lorea/comun#
```

Cambiar index.html por index1.html

```
root@serverapache:/Lorea/comun# mv index.html index1.html
root@serverapache:/Lorea/comun# ls
Fitxategi.txt  Fitxero2.txt  index1.html
root@serverapache:/Lorea/comun#
```

Systemctl restart apache2

Prueba desde Desktop:



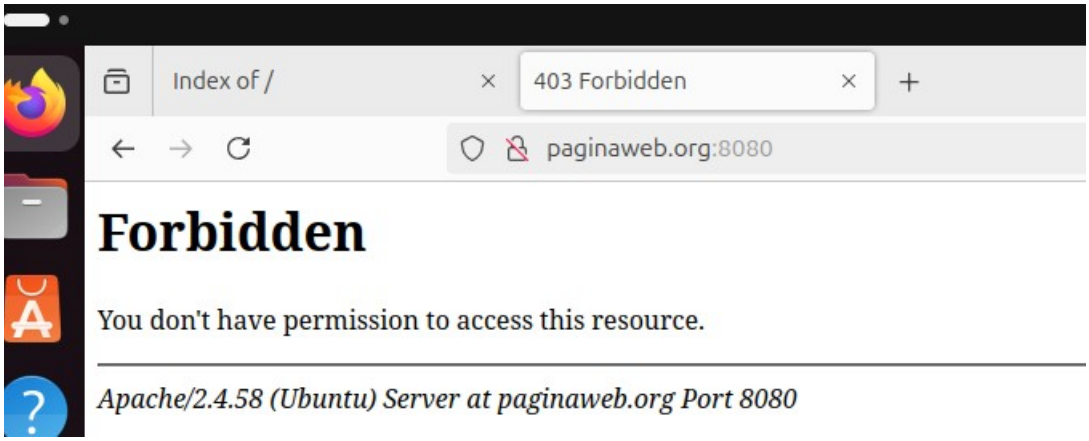
9. Prueba la funcionalidad de la directiva **Require ip** en el virtual host común para permitir el acceso solamente desde el Ubuntu Desktop.

nano /etc/apache2/apache2.conf

```
Directory /Lorea/>
        Options Indexes FollowSymLinks
        AllowOverride None
        Require all granted
/Directory>

Directory /Lorea/comun>
        Require ip 192.168.141
/Directory>
```

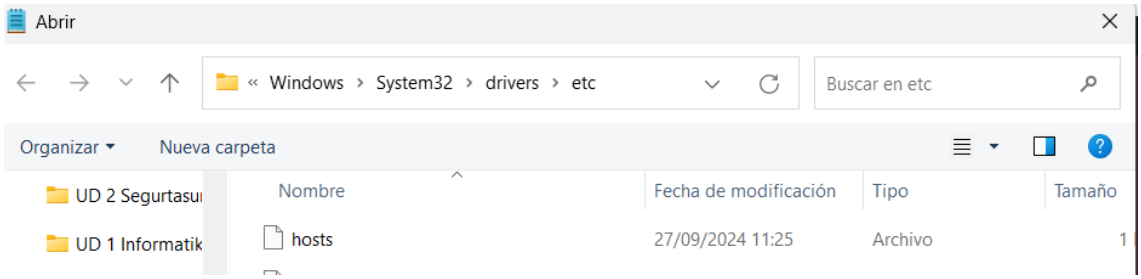
Systemctl restart apache2
Systemctl status apache2



OHARRA:

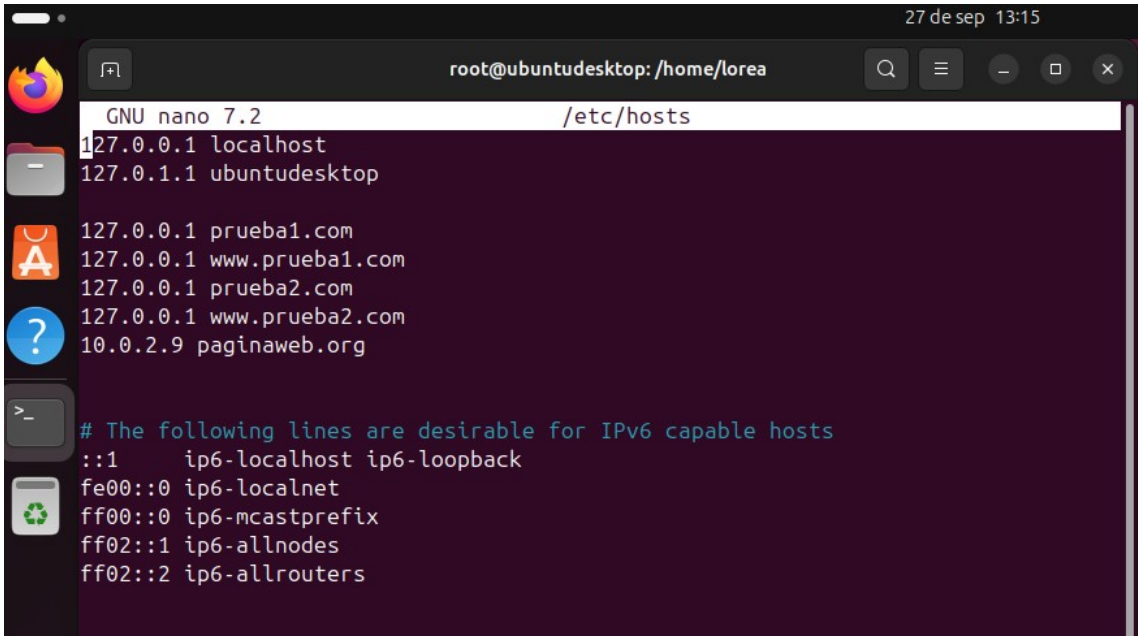
Gogoratu bai Desktop eta Windows-en hosts-en aldaketa egitea:

Windows-en (gogoratu administratzaile bezala sartzea)




```
192.168.1.136 prueba1.com www.prueba1.com
192.168.1.136 prueba2.com www.prueba2.com
192.168.141.3 paginaweb.org
```

Desktop


27 de sep 13:15
root@ubuntudesktop: /home/lorea
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 ubuntudesktop
127.0.0.1 prueba1.com
127.0.0.1 www.prueba1.com
127.0.0.1 prueba2.com
127.0.0.1 www.prueba2.com
10.0.2.9 paginaweb.org
The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters